Finding exposed companies with encoder models

How BERT-like models and related models help identify companies that are **active in a field** (a theme such as green hydrogen) or **exposed to a risk** (such as water scarcity or the EU carbon price), and how that fits ARP.

Status: concept note, 2026-11-01. Nothing here has been measured on ARP data yet; numbers marked *illustrative* are examples, not results. It belongs to the approved umbrella spec `docs/superpowers/specs/2026-11-01-model-stack-design.md` (roadmap phase 3, “Universe / Taxonomy”).

In one paragraph

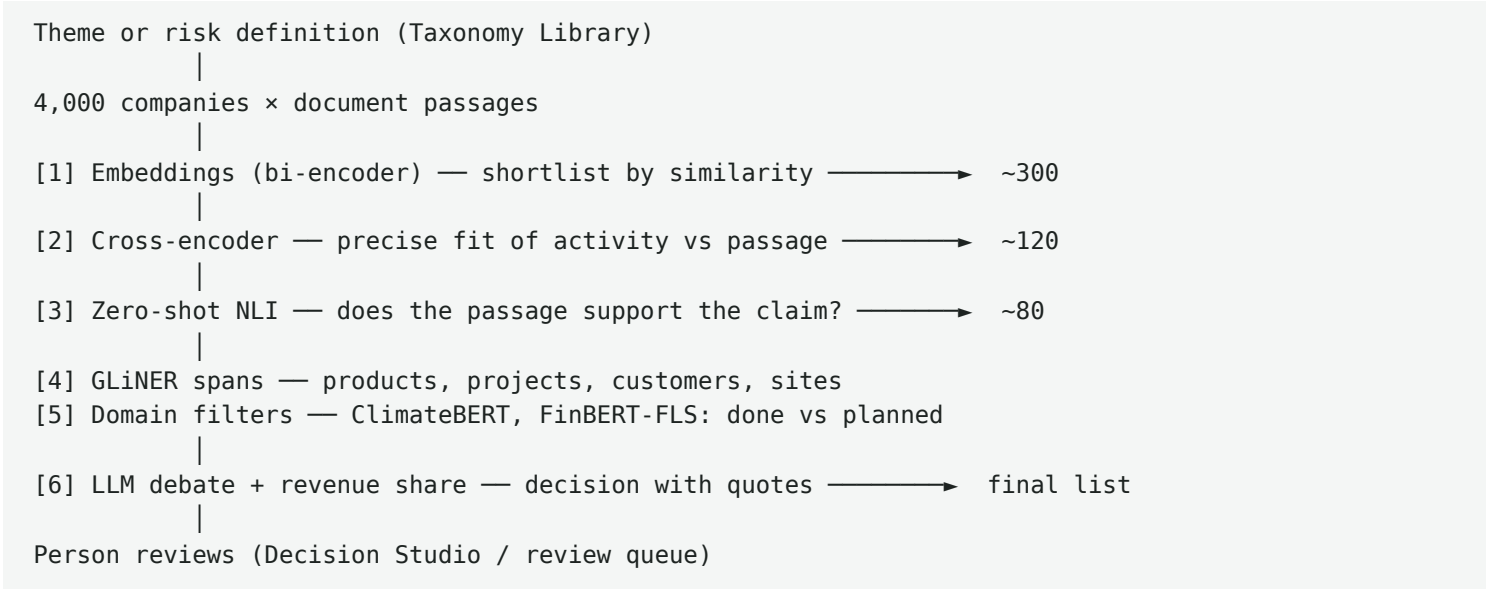
The models work as a **funnel**. Cheap models narrow about 4,000 companies to a shortlist, more precise models check the shortlist, and the LLM debate in the Thematic Universe Builder (function #1) decides only on what is left, citing exact quotes. Each step reads text and scores how well it fits the theme. No step can judge on its own *how much* of a company’s business that is: that needs revenue data.

1. Two questions, one method

Question	Example	Main text sources	Non-text sources
Activity: does the company do X?	“Makes electrolysers”, “Operates data centres”	business description (10-K Item 1, annual report “Business model”), segment notes, product pages, press releases	segment revenue (XBRL), patents, project databases
Risk exposure: is the company affected by Y?	“Water scarcity”, “EU carbon price”, “Asbestos litigation”	risk sections (10-K Item 1A, risk report, ESRS materiality), MD&A, earnings calls	plant locations × hazard maps, supply-chain graph, regulation scope (EU ETS installations)

The funnel is the same for both. Only the theme definition and the source documents change.

2. The funnel



Company counts are illustrative.

Step 0: define the theme (no model)

The Taxonomy Library (#2) holds the theme as a set of activities, each with a short description in plain words. Good descriptions matter more than the model: the models compare passages with these sentences.

Example, green hydrogen: “Manufactures electrolysers or electrolyser stacks”; “Produces hydrogen by electrolysis with renewable power”; “Builds or operates hydrogen pipelines or storage”; “Supplies components (compressors, valves, membranes) for hydrogen systems”.

Keep **producer** and **supplier** activities apart from the start; step 2 can only separate what the taxonomy separates.

Step 1: shortlist with embeddings

How it works: an embedding model turns each text into a vector, a list of a few hundred numbers that encodes its meaning. Texts with similar meaning get vectors that point in similar directions (high cosine similarity), even without shared words. All company passages are embedded once and stored; each activity description is embedded and compared with all of them in milliseconds.

Example: “We delivered 200 MW of PEM stacks to European customers” is close to “Manufactures electrolysers”, although the word “electrolyser” never appears.

Models: today `paraphrase-multilingual-MiniLM-L12-v2` (in `arp/retrieval/embeddings.py`); candidates `bge-m3`, `multilingual-e5-large`.

Setting: keep the top *k* passages per activity and every company with at least one passage above a similarity threshold. Set the threshold for **high recall**: a company dropped here never comes back.

Step 2: sharpen with a cross-encoder

How it works: the cross-encoder reads the activity and the passage *together* and outputs one fit score. It sees how the words relate, so it is more precise than step 1, but about 100× slower, so it runs only on the shortlist.

Example: “We supply valves to hydrogen projects” scores high for “Supplies components”, low for “Manufactures electrolysers”.

Models: `bge-reranker-v2-m3` (multilingual), `ms-marco-MiniLM-L-6-v2` (English, fast). ARP already has a reranker in `arp/retrieval/rerank.py` and on the `arp-models` service.

Step 3: is it real? (zero-shot NLI)

How it works: a natural-language-inference model answers whether a passage **supports**, **contradicts** or **says nothing about** a statement. Turn each activity into a statement (“The company makes or sells electrolysers”) and keep companies with at least one supporting passage. No training data is needed.

Example:

Passage	Verdict
“Order intake for electrolysers rose 40% to €310m.”	supports
“We are evaluating opportunities in hydrogen.”	neither (talk, not business)
“We exited the electrolyser business in 2024.”	contradicts

Models: `MoritzLaurer/deberta-v3-large-zeroshot-v2.0`, multilingual `mDeBERTa-v3-base-xnli`.

Step 4: what exactly? (GLiNER spans)

How it works: GLiNER marks text spans for labels you name at run time, without training. It produces the evidence a reviewer wants to see.

Example: “Our 100 MW electrolyser in Hamburg, built with Shell, starts in 2026” → product “electrolyser”, project “100 MW … Hamburg”, partner “Shell”, year “2026”.

Step 5: done vs planned (domain filters)

How it works: small fine-tuned classifiers separate actions from intentions and specific statements from boilerplate.

Model	Separates	Example
ClimateBERT commitment	action / commitment vs none	“commissioned a 20 MW unit” → yes
ClimateBERT specificity	specific vs vague	“€1.1bn for decarbonisation in 2025” → specific
FinBERT-FLS	forward-looking vs not	“will invest by 2030” → forward-looking

Step 6: decide (LLM debate + numbers)

The existing Universe Builder runs an Advocate, an Opposing agent and an Adjudicator over the evidence, each citing quotes that `arp/grounding.py` re-verifies. The **size** of the exposure comes from numbers, not from similarity: segment revenue from XBRL (#18) or the financials extraction (#4).

Example: “Electrolysers ≈ 35% of revenue (segment report, p. 112)” → core exposure; “Hydrogen projects < 1% of revenue” → peripheral.

3. Risk exposure: what changes

Step	Activity (theme)	Risk exposure
0	Activities	Risk statements: “The company is affected by water scarcity at its sites”
Sources	Business description, segments	Risk sections, ESRS materiality, MD&A, earnings calls
3 (NLI)	“The company makes X”	“The company says it is affected by Y”
6 (size)	Revenue share	Share of sites, capacity or revenue in affected regions; cost estimates the company gives

Example: Heidelberg Materials FY2025 lists water consumption as a financial risk (“Financial effects due to water scarcity and environmental regulations”), so steps 1–3 find it. Martin Marietta’s 10-K mentions drought only in passing; text alone would rank it lower.

Exposure companies do not write about needs non-text data:

- plant or asset locations crossed with hazard maps (water stress, flood, heat);
- the supply-chain graph (`arp/company_graph.py`) and input-output propagation (Indirect Exposure, #7);
- regulation scope lists (EU ETS installation register, CBAM sectors).

Text models only see what companies choose to disclose. Use both and flag the gap: “exposed by data, silent in text” is itself an engagement question.

4. Limits and how each is handled

Limit	Example	Handled by
Mentioning ≠ doing	“Hydrogen could play a role” scores like a product line	step 3 (NLI), step 5 (forward-looking filter)
Relevant ≠ material	A conglomerate with 0.5% hydrogen revenue looks as close as a pure player	step 6, revenue share
Supplier ≠ producer	Valve maker vs electrolyser maker	separate activities (step 0) + cross-encoder (step 2)
Old or missing text	New products not yet in last year’s report	Document Discovery (#6), news, earnings calls
Language	German or French reports	multilingual models (bge-m3, mDeBERTa) or translation first
Boilerplate	Every report says “climate change may affect us”	specificity filter (step 5), peer comparison
Thresholds	Which score counts as “exposed”?	set on a golden set, see section 5

5. Setting thresholds and measuring quality

1. **Golden set:** about 50–100 companies per theme, hand-labelled as core / peripheral / not exposed, mixed by sector, region and language, with the hard cases (suppliers, announcers, divested).
2. **Recall first in steps 1–2.** Set the thresholds so ≥ 95% of truly exposed companies survive; precision comes later.
3. **Precision at the end.** Measure the share of companies in the final list that the labels confirm.
4. **Compare with and without the funnel** (umbrella-spec rule): same final quality with fewer LLM calls, or better quality at the same cost.
5. **Re-check per theme.** A threshold tuned for hydrogen does not carry over to water risk.

6. Cost and run time

For 4,000 companies × about 200 passages each (about 800k passages), *estimates, not benchmarks*:

Step	Runs on	Time	Compute cost
1 Embeddings	L4 GPU (CPU possible)	~15 min GPU, a few hours CPU; once per document, reused across themes	< $1
2 Cross-encoder	L4	~300 companies × 20 passages × activities: minutes	< $1
3 NLI	L4	minutes on the shortlist	< $1
4–5 GLiNER, domain filters	L4	minutes	< $1
6 LLM debate	Claude API	only on ~80 companies instead of 4,000	the main saving

The embeddings are computed once per document and serve every theme and risk, which is what makes the funnel cheap.

7. Where this lives in ARP

Piece	Today	Change
Theme definitions	Taxonomy Library (#2)	none; producer/supplier split is a content task
Embeddings, reranker	`arp/retrieval/`, `arp-models` service	move behind `arp/models/` (umbrella spec)
NLI, GLiNER, domain classifiers	not present	`classify`, `spans`, `tag` in `arp/models/` (umbrella spec)
LLM debate	`arp/research/matcher_agents.py`, `match_graph.py`	receives the shortlist instead of every candidate
Revenue share	XBRL Facts (#18), Financials (#4)	none
Non-text exposure	`arp/company_graph.py`, Indirect Exposure (#7)	none
Review	Decision Studio, review queue	none

Implementation is a child spec of the umbrella spec, behind a setting that is off by default, and becomes the default only if the golden-set comparison in section 5 shows a gain.

8. Open questions

- Which theme to use for the first golden set (a well-defined one such as electrolysers, or a fuzzy one such as “AI infrastructure”)?
- Passage source: annual reports only, or also product web pages and press releases (fresher, noisier)?
- Should “exposed by data, silent in text” feed engagement selection directly?